The outline currently includes crisis leadership, adaptive leadership, transformational leadership, distributed leadership, digital leadership, and learning-oriented leadership. This broad coverage could dilute the central argument.

Based on your search evidence, you need to show me whether antifragility is the primary construct being examined or simply one of several possible responses to disruption. The introduction should explain why resolving these gaps changes leadership theory.

Develop the argument around four issues:

1. Conceptual ambiguity: Does antifragility describe a distinct leadership capability, an organizational property, or an outcome produced by leadership practices?
2. Theoretical differentiation: How does antifragility differ from resilience, adaptive leadership, organizational learning, and dynamic capabilities?
3. Causal mechanisms: Under what conditions can leadership convert disruption into learning, innovation, and improved performance?
4. Boundary conditions: When does disruption stimulate improvement, and when does it deplete resources, increase vulnerability, or undermine performance?

The introduction should also distinguish among three levels of analysis:

- Individual: leader resilience, decision-making and learning.
- Team: collective adaptation, psychological safety and shared learning.
- Organization: recovery, innovation, capability development and performance.

These levels are related but should not be treated as interchangeable. For example, an employee's perception of resilient leadership is not equivalent to an objectively measured organizational capacity to improve following a crisis.

You can write something as suggested below:

The review seeks to clarify whether antifragility represents a distinct leadership construct or an organizational outcome enabled by particular leadership practices. By synthesizing the conceptual, empirical, and quantitative evidence, it will identify the mechanisms and contextual conditions through which leadership supports recovery, adaptation, and improvement following disruption.

This contribution is stronger because it poses a question that the review can investigate rather than presuming the answer.

Make the research questions more analytical:

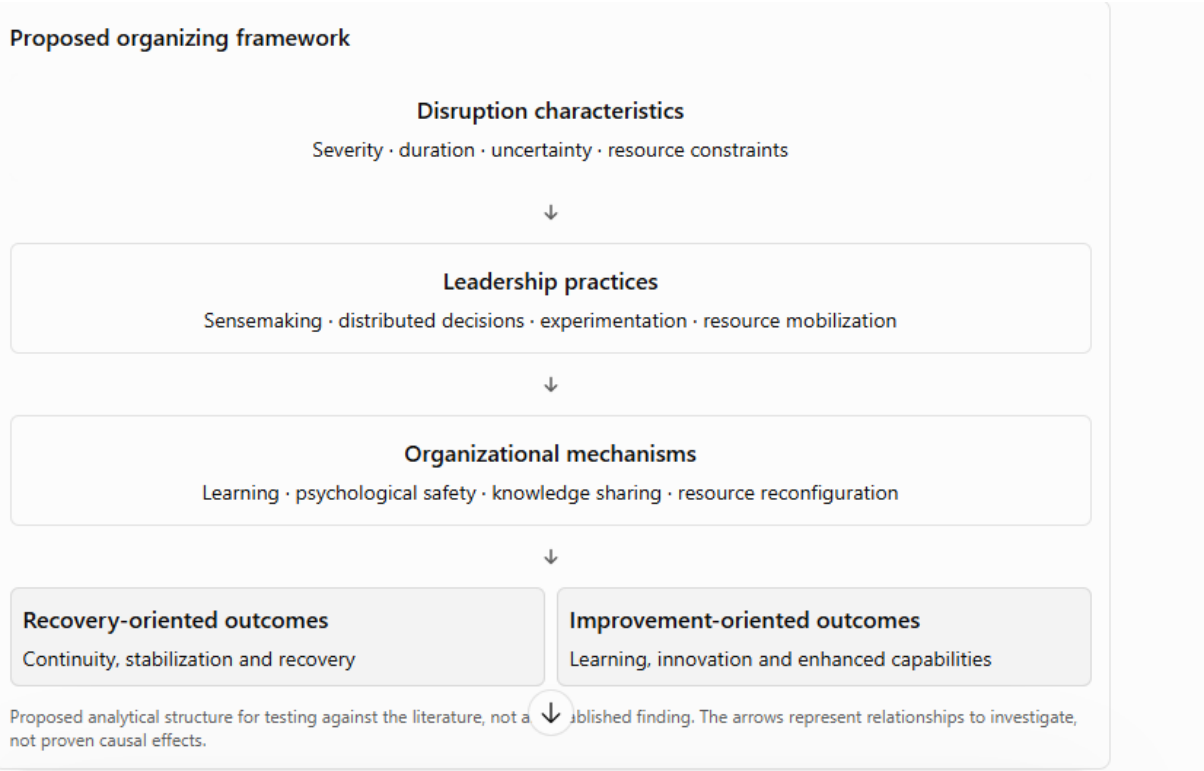
The six existing research questions cover conceptualization, leadership approaches, mechanisms, contextual factors, quantitative relationships, and improvement through disruption. Their coverage is reasonable, but RQ2, RQ3, and RQ6 overlap. I suggest consolidating them into five questions.

Revised research questions	Purpose
RQ1. How are resilient leadership and antifragile leadership conceptualized, distinguished, and operationalized in existing research?	Construct clarification
RQ2. Through which leadership practices and organizational mechanisms are disruption-related outcomes influenced?	Explanation of mechanisms
RQ3. What empirical relationships exist between leadership-related constructs and organizational adaptation, learning, innovation, and performance?	Evidence synthesis and meta-analysis
RQ4. How do contextual conditions, levels of analysis, and methodological differences shape these relationships?	Boundary conditions and moderators
RQ5. To what extent does existing evidence support the proposition that organizations can improve through disruption, beyond recovering or maintaining performance?	Central theoretical contribution

One important point: RQ3 should be divided into specific quantitative hypotheses or planned associations in the methods protocol. A meta-analysis cannot reliably pool every leadership-organizational outcome relationship into one overall effect.

Suggestion on Theoretical Framework

The outline discusses multiple leadership approaches but does not yet specify how these will be integrated into a coherent theoretical explanation. Select a primary theoretical lens and use complementary theories only where they explain a specific mechanism.



Use this framework as a provisional coding structure, not as something imposed on the findings. The final model may need to be modified or rejected depending on the evidence.

The theoretical foundations could be organized as follows:

- Organizational resilience: explains the capacity to anticipate, absorb, respond to, and recover from adversity.
- Dynamic capabilities: helps explain how sensing, seizing and reconfiguring resources may support adaptation and transformation.
- Organizational learning: explains how experience, experimentation and feedback may generate knowledge and capability development.
- Antifragility: provides the proposition that certain systems can benefit from variability or stressors, a proposition that requires careful operational and empirical scrutiny.

A key distinction is that resilience and antifragility should not automatically be presented as successive stages. An organization may recover successfully without improving, and an organization may learn from disruption without every outcome improving. The review should investigate these possibilities explicitly. The methodology on pages 4–10 is a good starting point, particularly its two complementary Scopus searches and its distinction between thematic synthesis and quantitative pooling.

Databases and search coverage

The outline proposes Scopus, PubMed, and Google Scholar. It also notes that PubMed will likely contribute healthcare research primarily and that Google Scholar requires a predefined screening limit.

Recommended changes:

- Retain Scopus as a core source.
- Consider adding Web of Science Core Collection and a specialist business database such as ABI/INFORM or Business Source Complete, subject to access.
- Retain PubMed if healthcare leadership and healthcare crises are within the defined scope.
- Use Google Scholar primarily for supplementary discovery and citation chasing, with a reproducible procedure for the records screened.
- Conduct backward and forward citation searching for included studies and relevant prior reviews.

Do not add databases merely to increase the number of sources. Each database should have a clear role, and document all searches.

Refine the search strings.

The existing search strategy covers leadership and disruption, as well as resilience and antifragility. However, the second search may miss studies that examine leadership and antifragility without explicitly mentioning disruption. The first search may also retrieve an excessively broad set of studies about leadership and organizational change.

Use three complementary search families:

Search family	Main purpose
Leadership and disruption	Capture crisis leadership and other leadership responses without requiring resilience terminology.
Leadership and resilience/antifragility	Identify studies directly examining the central constructs.
Leadership, learning and positive adaptation	Capture studies examining improvement through learning, experimentation and capability development using different terminology.

Test each search against a small set of known relevant studies. This is a useful way to identify whether the strategy is too restrictive. The final methods section must include the exact database-specific search strings, search dates, filters, and retrieval counts. Test the strings in the actual databases rather than presenting them only as proposed queries.

Define the review boundaries.

The current eligibility criteria allow quantitative, qualitative, mixed-methods, and conceptual studies across business, management, public-sector, and healthcare settings.

That breadth suits some forms of conceptual synthesis, but it creates a challenge for quantitative pooling.

Specify:

- Whether the primary unit is leadership behavior, leadership capability, leader characteristics, or organizational leadership practices.
- Which levels of analysis are eligible.
- Whether organizational resilience is an outcome, a mediator, or a predictor in each relationship.
- Which disruption types qualify.
- Whether cross-sectional studies are eligible and how their findings will be interpreted.
- Whether conceptual papers will be included in theory development but excluded from empirical effect-size calculations.
- Which publication types, languages, and date ranges are eligible, with justification.

I recommend keeping the thematic review broad enough to capture conceptual diversity while defining narrower, prespecified evidence groups for meta-analysis.

Screening and data extraction:

The outline appropriately recommends independent screening by two reviewers. This should be strengthened by specifying:

- Independent title/abstract screening and full-text screening.
- A documented process for resolving disagreements.
- Pilot testing of the screening criteria.
- A standardized extraction form, tested on a sample of studies.
- Extraction of adjusted and unadjusted estimates, measurement reliability, and relevant design characteristics.
- Procedures for missing statistics, multiple publications from the same sample and multiple effect sizes from a single study.

For transparency, maintain a record of exclusion reasons at the full-text stage and report the final study selection in the PRISMA 2020 flow diagram.

Quality appraisal and confidence in evidence:

The outline proposes JBI or MMAT tools and RoB 2 if randomized trials are included. The final protocol should select the appraisal tool according to the designs actually eligible for the review. A menu of tools is not a complete appraisal plan. In particular:

- Use design-appropriate criteria for observational, qualitative and mixed-methods studies.
- Explain how appraisal results will influence the synthesis and interpretation.
- Avoid automatically excluding studies solely because they receive a low quality score unless exclusion rules are justified in advance.
- For each meta-analysis, consider how risk of bias, inconsistency, indirectness and imprecision affect confidence in the pooled estimate.

If the evidence permits, a structured assessment of certainty for each major quantitative finding will make the conclusions more useful to both scholars and practitioners.

Meta -analyses:

The current outline proposes Fisher's \(z\) transformation for correlations, standardized mean differences for continuous outcomes, risk ratios or odds ratios for binary outcomes, random-effects models, heterogeneity analysis, and sensitivity tests. These are reasonable candidate methods, but they do not yet establish what the meta-analysis will actually estimate.

The Primary relationships to be specified before screening:

Identify a small number of priority relationships, subject to confirmation that sufficient comparable studies exist.

Proposed relationship	Potential synthesis
Leadership practices → organizational resilience.	Pooled correlations or comparable adjusted effects
Leadership practices → organizational learning.	Pooled correlations or comparable effects
Leadership practices → innovation and adaptation.	Separate outcome-specific analyses
Organizational resilience → performance	Separate synthesis if construct definitions are sufficiently comparable
Leadership practices → antifragility-related outcomes	Exploratory synthesis, conditional on adequate empirical evidence

These are proposed analyses, not claims that sufficient evidence already exists. Do not assume that every relationship will support a meta-analysis.

About the statistical plan:

The protocol should specify the following elements.

1. Effect-size selection

Identify the preferred effect size for each outcome and explain how other statistics will be converted. Correlations, standardized mean differences and odds ratios should not be combined indiscriminately.

2. Random-effects model

A random-effects model may be appropriate because studies are likely to differ in populations, settings and measures. Specify the estimator for between-study variance, confidence-interval method and any small-sample corrections as appropriate to the data.

3. Dependence between effect sizes

One study may report several leadership measures, outcomes, or samples. These effect sizes are not necessarily independent. Specify whether the review will select one effect per study, aggregate compatible measures, use multilevel meta-analysis, or apply robust variance estimation where justified.

Heterogeneity and moderators

Report \(I^2\), \(\tau^2\) and Cochran's \(Q\), as appropriate. Plan moderator analyses only where sufficient studies and relevant variation are available. Potential moderators include disruption severity, sector, study design, level of analysis, and leadership approach.

5. Publication bias and sensitivity

Use funnel plots and statistical tests only when the number and characteristics of studies make them informative. Sensitivity analyses should examine the influence of high-risk-of-bias studies, influential observations and alternative defensible analytical decisions.

6. Reporting null findings

Report the direction, magnitude and uncertainty of findings, including null and inconsistent results. Do not interpret statistical significance as a direct measure of managerial importance.

If there is limited antifragility research:

This is an important possibility because the central term may be used inconsistently or may not have been operationalized in empirical leadership research. If direct evidence is insufficient, the manuscript can still make a valuable contribution through:

- A systematic conceptual synthesis of antifragility in leadership.
- A quantitative synthesis of better-established, related constructs where pooling is justified.
- An evidence map showing which relationships have been tested and which remain theoretical.
- A transparent explanation of why antifragile leadership cannot yet be evaluated quantitatively as a distinct construct.

This approach is more credible than treating resilience, adaptability, learning, and antifragility as interchangeable measures.

Results and Discussion: move from classification to explanation:

The outline's results section covers study characteristics, conceptualizations, leadership responses, mechanisms, outcomes, quantitative synthesis, and methodological quality. The discussion then revisits the resilience–antifragility distinction and identifies theoretical and practical implications. The proposed coverage is comprehensive, but the manuscript risks repeating the same categories across several sections.

Restructure the findings around three substantive questions.

Finding 1: What do the evidence and the literature mean by resilience and antifragility?

Establish construct definitions, levels of analysis, measurement approaches, and conceptual overlaps. Distinguish established empirical constructs from theoretical propositions.

Finding 2: How might leadership enable improvement through disruption?

Organize the evidence around learning, experimentation, resource reconfiguration, participation, and other supported mechanisms. Examine contradictory findings and contextual conditions.

Finding 3: Which relationships are empirically supported?

Present the pooled estimates where feasible, alongside uncertainty, heterogeneity, evidence quality, and the limitations of causal interpretation. Separate these findings from narrative and conceptual conclusions.

The discussion should then address the implications of these findings for leadership theory, organizational adaptation and the development of antifragility as a construct.

Before finalizing the outline, I recommend adding the following deliverables.

Pre-submission readiness checklist 7 of 7

- **Document the novelty of the review**
  Compare existing systematic reviews and meta-analyses on crisis leadership, organizational resilience, adaptive leadership and antifragility. Show what remains unanswered.
- **Finalize and document the protocol**
  Specify eligibility rules, search procedures, primary outcomes, planned effect sizes and decision rules before conducting the full review.
- **Pilot the search and screening**
  Test whether known relevant studies are retrieved and whether reviewers apply the eligibility criteria consistently.
- **Build a construct and outcome codebook**
  Distinguish leadership inputs, mediating mechanisms, contextual moderators and outcomes at individual, team and organizational levels.
- **Conduct a feasibility assessment**
  Review an initial sample of eligible quantitative studies to determine whether compatible effect sizes and outcomes are available.
- **Plan the evidence presentation**
  Prepare templates for the PRISMA flow diagram, study characteristics, construct comparison, quality appraisal and forest plots where applicable.
- **Align conclusions with evidence**
  Separate empirical findings from conceptual propositions and avoid causal claims based solely on observational associations.